Minn Bo Bo

Software Engineer & Researcher | Software & Distributed Systems

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RESEARCH PROFILE

Software engineer and Electrical and Computer Engineering researcher with an MEng GPA of 4.00/4.00. Builds production systems across frontend, backend, data, cloud infrastructure, DevOps, and real-time communication. Primary developer of three integrated MANGOs platforms and first author of two IEEE publications on identity, interoperability, and tokenomics for interconnected metaverses.

Research interests: Distributed software systems; cloud-native architecture; secure identity and interoperability; scalable web platforms; software architecture; DevOps; blockchain ecosystems; reliability and performance engineering.

TECHNICAL SKILLS

Backend: Node.js, TypeScript, Express.js, Flask, REST APIs, OAuth/SSO, WebSockets, Sequelize ORM, Azure Functions

Frontend: React, React Native, Redux Toolkit, JavaScript, HTML, CSS, SCSS

Data: PostgreSQL, MongoDB, Microsoft SQL Server, InterSystems Caché, SQL, data modelling

Cloud & DevOps: AWS, EC2, IAM, S3, Route 53, CloudWatch, Docker, Docker Compose, Nginx, Linux, GitHub Actions, CI/CD

Quality: Jest, unit testing, integration testing, API validation, deployment and recovery practices

Research & AI: Python, C++, TensorFlow, TensorFlow.js, blockchain/tokenomics, document processing

PROFESSIONAL EXPERIENCE

Software Engineer - Full Stack & DevOps
Aug 2024 - Present

Terrafirma | Software delivery for William Green, Australia

- Develop and maintain production full-stack systems for an Australian dental equipment distribution and servicing company.
- Build TypeScript and Node.js backend services, REST APIs, and PostgreSQL data workflows.
- Support delivery with AWS, Docker Compose, Nginx, Linux, GitHub Actions, CI/CD, Jest, unit testing, and integration testing.

Full-Stack Developer & DevOps Engineer
Oct 2023 - Aug 2024

iBeky | Medical services technology

- Led Node.js backend development for internal web and mobile projects using Express.js and Jest.
- Provisioned AWS environments, configured Nginx reverse proxies, and administered databases and disaster-response procedures.
- Built Python utilities and Flask applications for AI-enabled document and data workflows.

Junior Full-Stack Developer
Jan 2023 - Oct 2023

Jonda Health | Singapore healthcare technology

- Served as principal backend developer for JondaX, guided interns, developed Python document-processing tools, and integrated AWS services.

Backend Developer & Database Designer
2021 - 2022

Australian learning management system startup

Designed database systems, secure routes, permission models, and REST-based Azure Functions with Node.js, Express.js, Sequelize ORM, and JavaScript.

EDUCATION

Master of Engineering, Electrical Engineering | GPA: 4.00/4.00 **2024 - 2026**
Chulalongkorn University, Thailand | Telecommunication and Networking Systems

BEng (Hons), Electrical and Electronic Engineering - First Class **2022 - 2024**
Liverpool John Moores University, United Kingdom

Electronic and Communication Engineering **2016 - 2022**
Yangon Technological University, Myanmar | Transferred

SELECTED RESEARCH & ENGINEERING SYSTEM

MANGOs Multi-Platform Software Ecosystem - Primary Developer
MANGOs | School | Scene Studio | Chulalongkorn University
Mentored by Professor Lunchakorn Wuttisittikulkij

- Architected, built, and continue to operate three integrated platforms, owning every application screen plus backend services, APIs, socket communication, data models, databases, storage, deployments, and infrastructure.
- Implemented centralized identity and OAuth/SSO, role-based administration, metaverse registries, avatars, social messaging, marketplace and wallet workflows, public API integrations, and organization-aware learning services.
- Built the School course-management and progress systems and the Scene Studio authoring and versioned-publishing workflows; established clean integration boundaries for course authors, 3D artists, and Unity-metaverse collaborators.

SELECTED PUBLICATIONS

1. **M. B. Bo** et al., "Authentication and Identification System for Interconnected Metaverses: Consideration and Implementation," IEEE ECTI-CON, 2025. doi:10.1109/ECTI-CON64996.2025.11100953
   Designed and implemented identity management and interoperability across participating virtual environments.
2. **M. B. Bo** et al., "Design and Implementation of Tokenomics and Ecosystem Model for Interconnected Metaverses," IEEE CCISP, 2025, pp. 225-229. doi:10.1109/CCISP67522.2025.11282105
   Designed and implemented an ecosystem framework for participation, incentives, and value exchange.

SELECTED AWARDS

- **Third Prize, Huawei Developer Competition Asia Pacific** (2024)
  Developed a blockchain-powered academic publishing platform on Huawei Cloud for research publishing, academic integrity, reputation, and revenue workflows. Competition result.
- **Merit Award, Asia Pacific ICT Alliance Awards** (2023)
  Led a two-stage deep learning system for lung segmentation and pulmonary tuberculosis detection from chest X-rays, integrated into a doctor-patient cloud ecosystem. Official winner list.
- **First Prize, APT IoT Smart Challenge for Agri-Tech** (2022)
  Developed AWS-backed cloud infrastructure and a deep learning component for an Arduino- and microcontroller-integrated smart greenhouse system.

CERTIFICATIONS & LANGUAGES

Certifications: DevOps on AWS Specialization; Deep Learning Specialization; DeepLearning.AI TensorFlow Developer Professional Certificate; TensorFlow: Advanced Techniques Specialization.

Languages: Burmese (native); English (IELTS 8, official test; Diploma in English, Wall Street English).